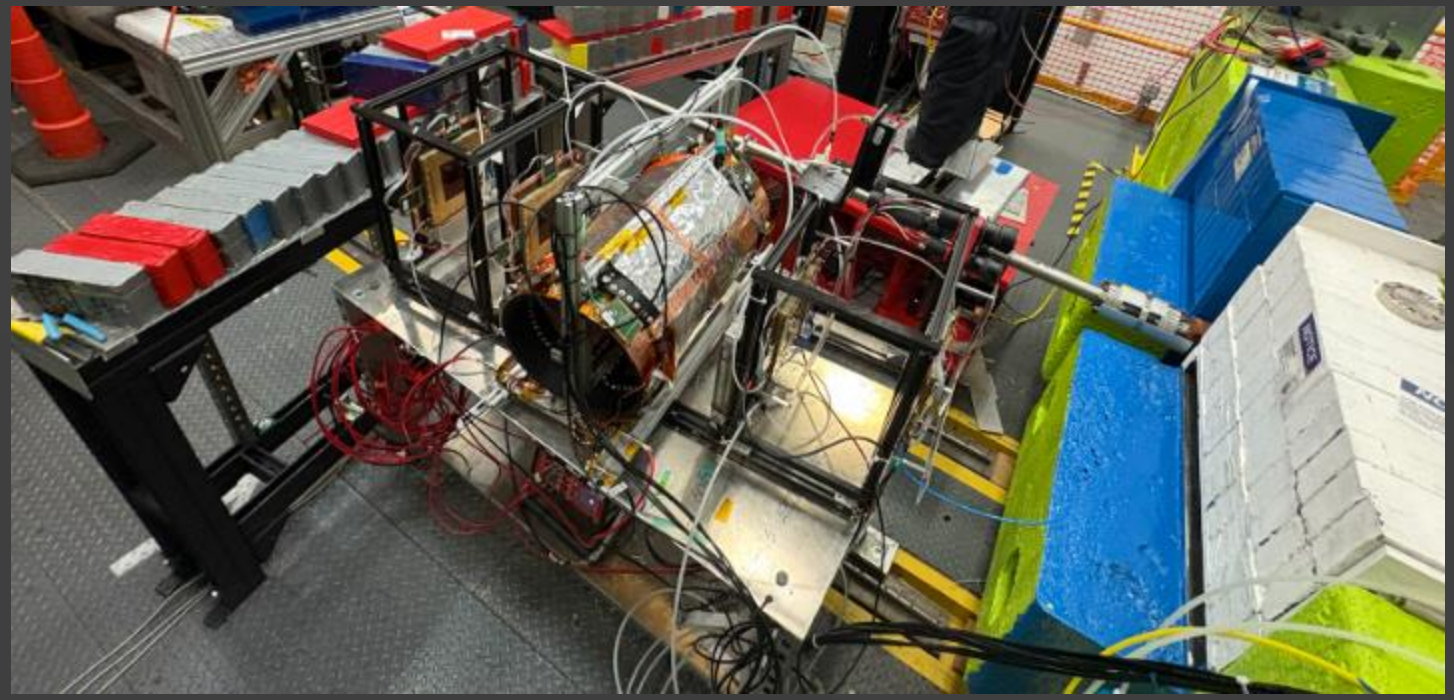

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Mapping for a Cylindrical Micro-Resistive Well Detector

Milestone 2

People

Group Members

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- Dr. Marcus Hohlmann, hohlmann@fit.edu

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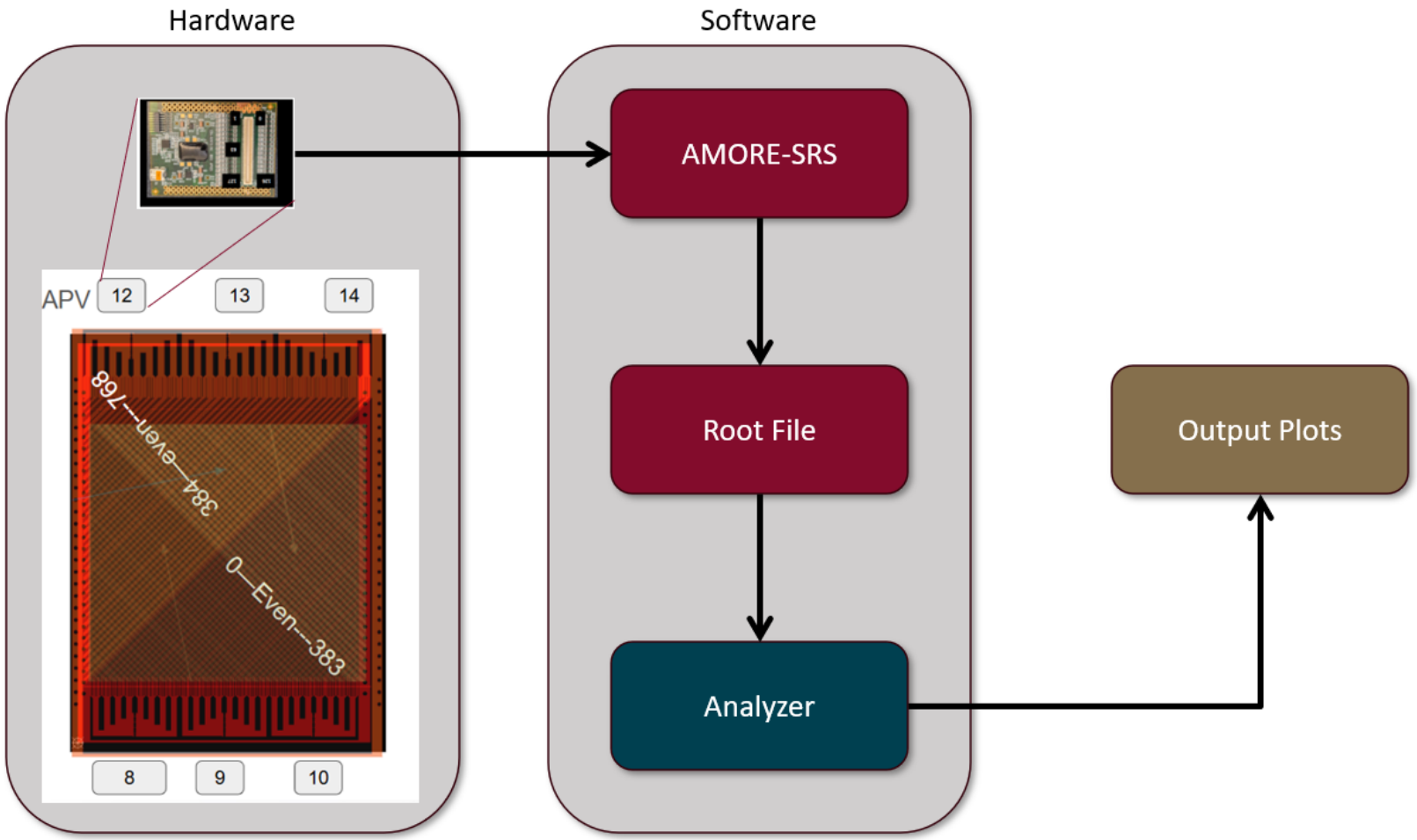
Overview



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Milestone 2 Task Matrix

Task	Completion	Alejandro	Summer	TO DO
1. Reconstruct hit position from the tracker data.	100%	0%	100%	Complete.
2. Generate resolution plots.	100%	100%	0%	Complete.
3. Investigate how root interacts with AMORE-SRS.	70%	10%	60%	We investigated the interaction between root and AMORE-SRS indirectly when we started the mapping for the cylindrical detector.

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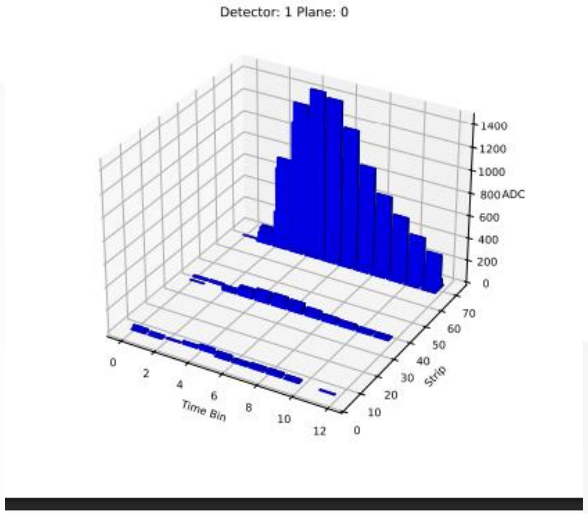
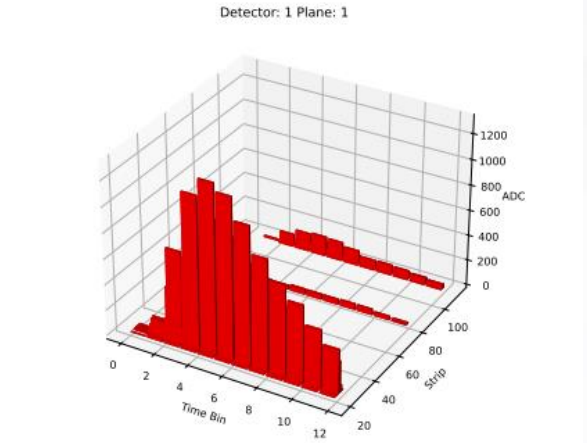
Task- Reconstruct hit positions





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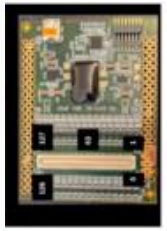
Task- Generate resolution plots

- Demo video: https://youtu.be/QcxFqzgQe2Q

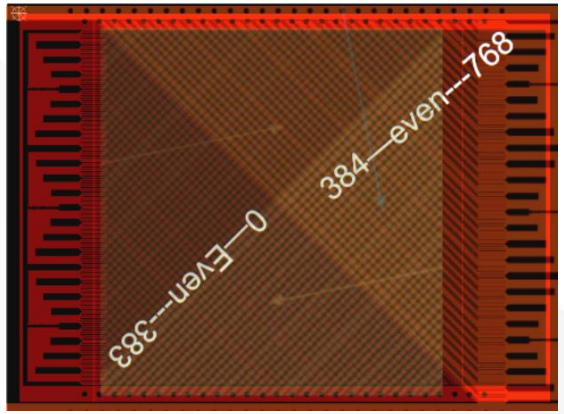
Task- Generate resolution plots

- Filter and preprocess the data of each event.
- Gathered good/useful events.
- Match the hits of an event into (x, y) pairs.
- Get the corresponding z value(adc) by calculating center of mass.
- Plot the resulting grid.

Task- Investigate AMORE-SRS representation



Planar Tracker




```
==============================
Detector 0
==============================
  Total hits:        486929
  Strip min:         0
  Strip max:         255
  Unique strips:     256
  First sample:      [ 30  31  32  61
 168 169]

==============================
Detector 1
==============================
  Total hits:        601468
  Strip min:         256
  Strip max:         510
  Unique strips:     254
  First sample:      [286 287 316 317
 451 452]

==============================
Detector 2
==============================
  Total hits:        549364
  Strip min:         512
  Strip max:         767
  Unique strips:     256
  First sample:      [540 543 546 575
 709 739]

==============================
Detector 3
==============================
  Total hits:        349666
  Strip min:         768
  Strip max:         1023
  Unique strips:     256
  First sample:      [ 799  806  825
  839  840  841  853  887  927]
```

```
==============================
Detector 4
==============================
  Total hits:        543908
  Strip min:         384
  Strip max:         1023
  Unique strips:     256
  First sample:      [402 428 434 438 440
 911 913]

==============================
Detector 5
==============================
  Total hits:        389920
  Strip min:         0
  Strip max:         895
  Unique strips:     768
  First sample:      [181 186 188 190 209
 188 224]
```

- Tracker 0 → strips **0–255**
- Tracker 1 → strips **256–511**
- Tracker 2 → strips **512–767**
- Tracker 3 → strips **768–1023**

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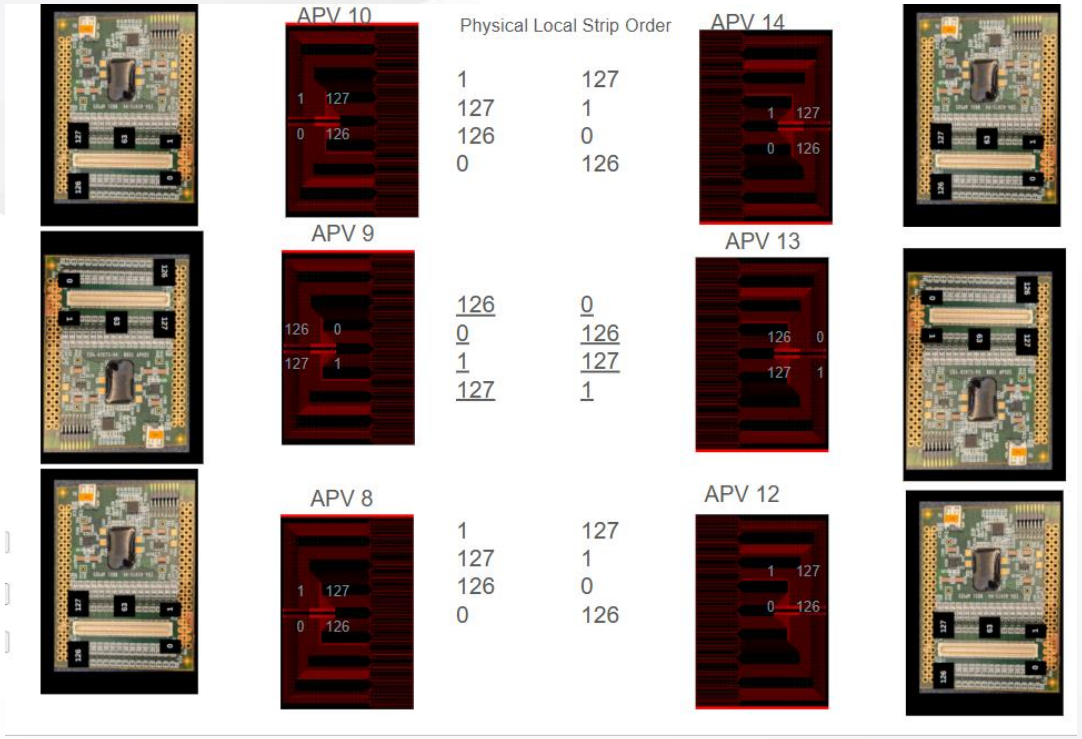
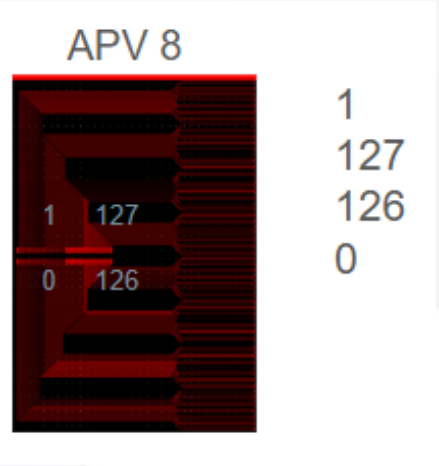
Milestone 3 Tasks

- Generate cylindrical mapping functions
- Resolution plots for cylindrical detector data
- Validate cylindrical plots against tracker plots



0	0
1	127
2	1
3	126
4	2
5	125
6	3
7	124
8	?
9	?
10	?

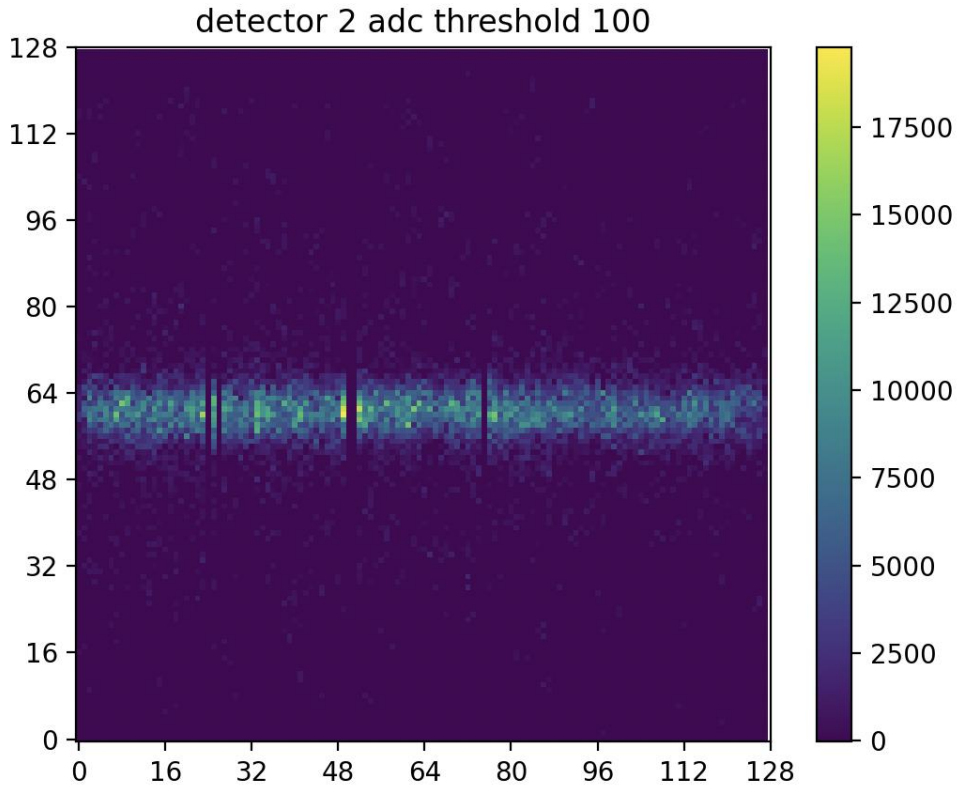


Milestone 3 Tasks

- Generate cylindrical mapping functions

➢ Resolution plots for cylindrical detector data

- Validate cylindrical plots against tracker plots



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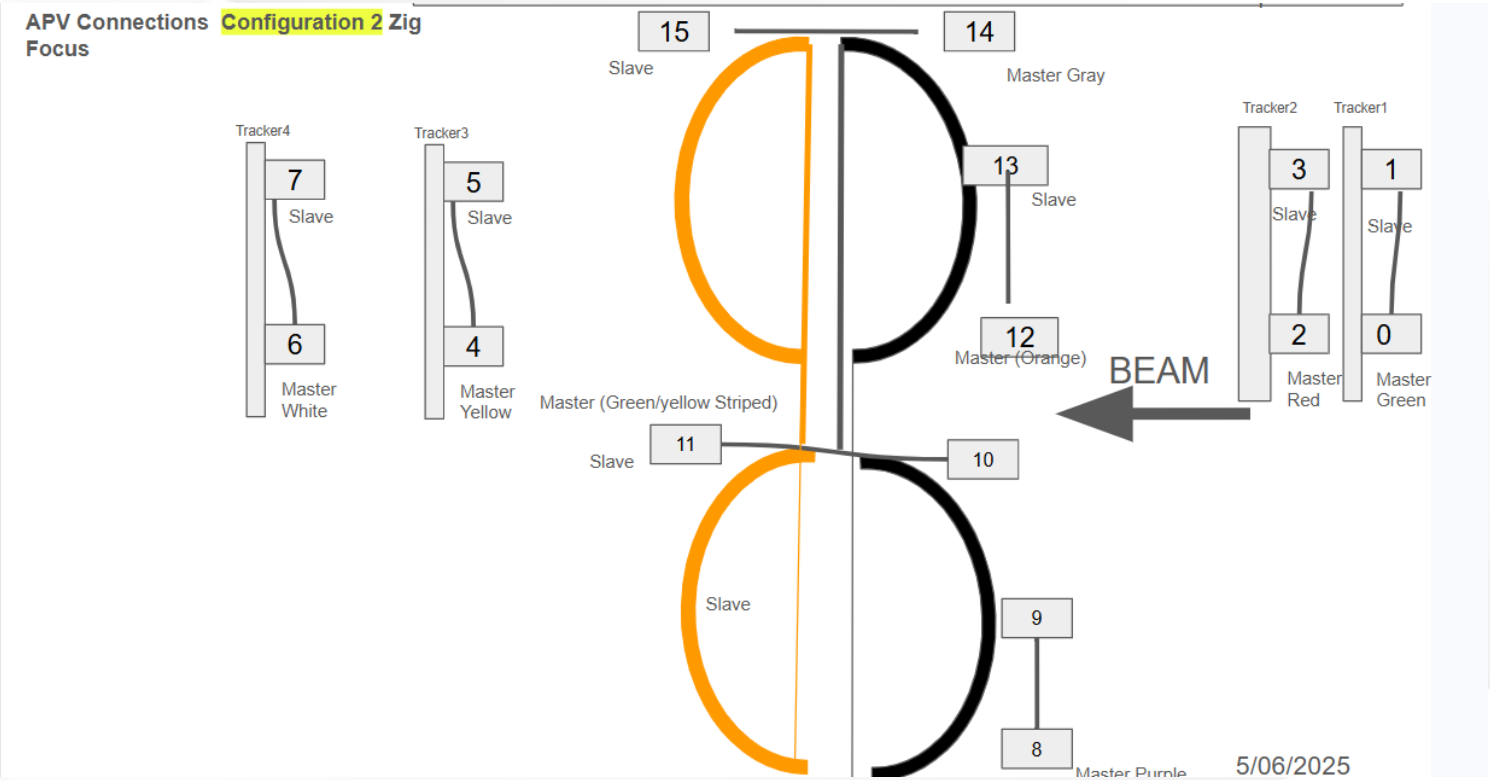
Milestone 3 Tasks

- Generate cylindrical mapping functions
- Resolution plots for cylindrical detector data

➢Validate cylindrical plots against tracker plots



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Milestone 3 Task Matrix

Task	Alejandro	Summer
1. Generate cylindrical mapping functions	50%	50%
2. Generate resolution plots for cylindrical detector	50%	50%
3. Validate cylindrical resolution plots against tracker resolution plots	50%	50%

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Thank you for listening

Questions?

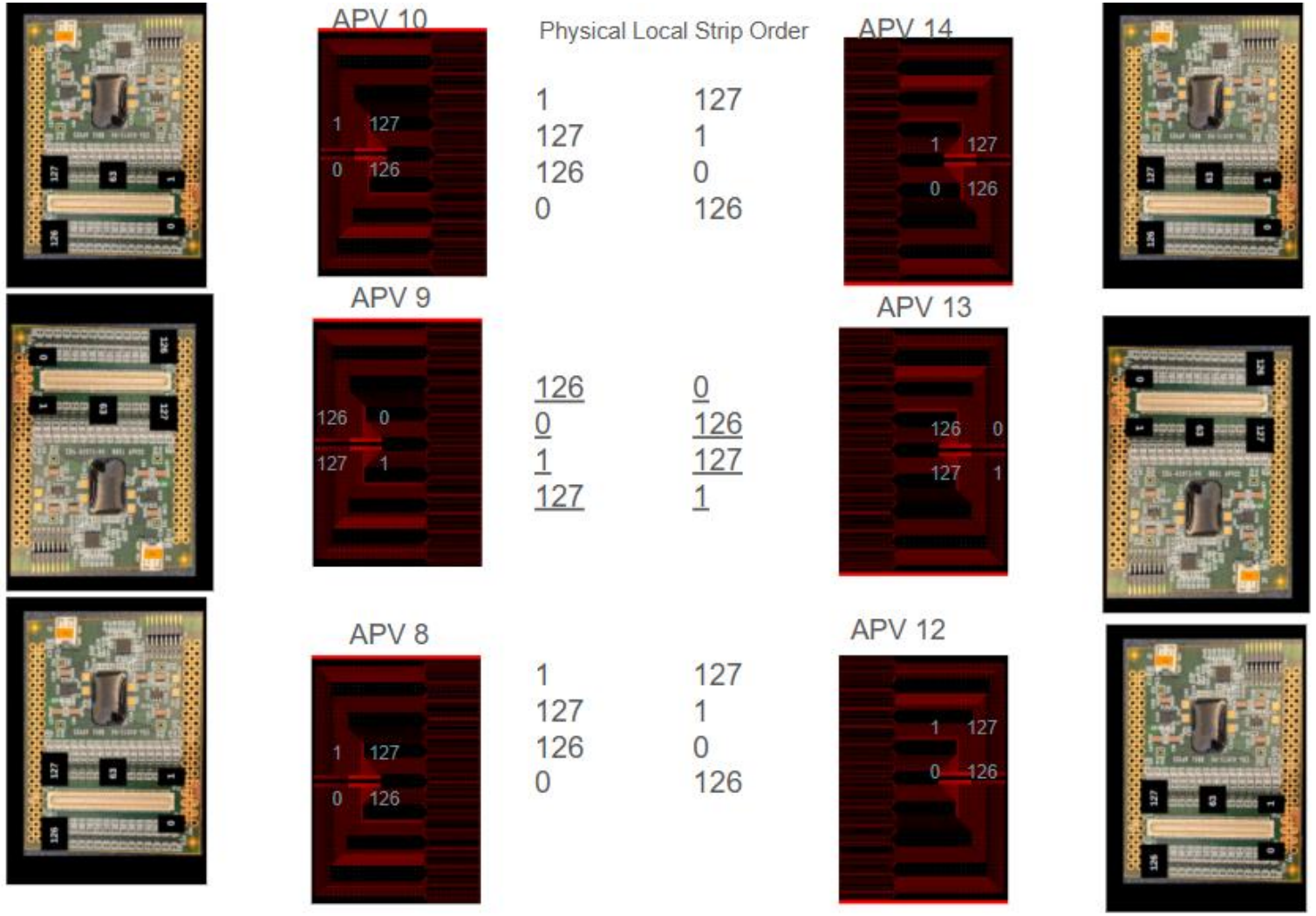
Definitions

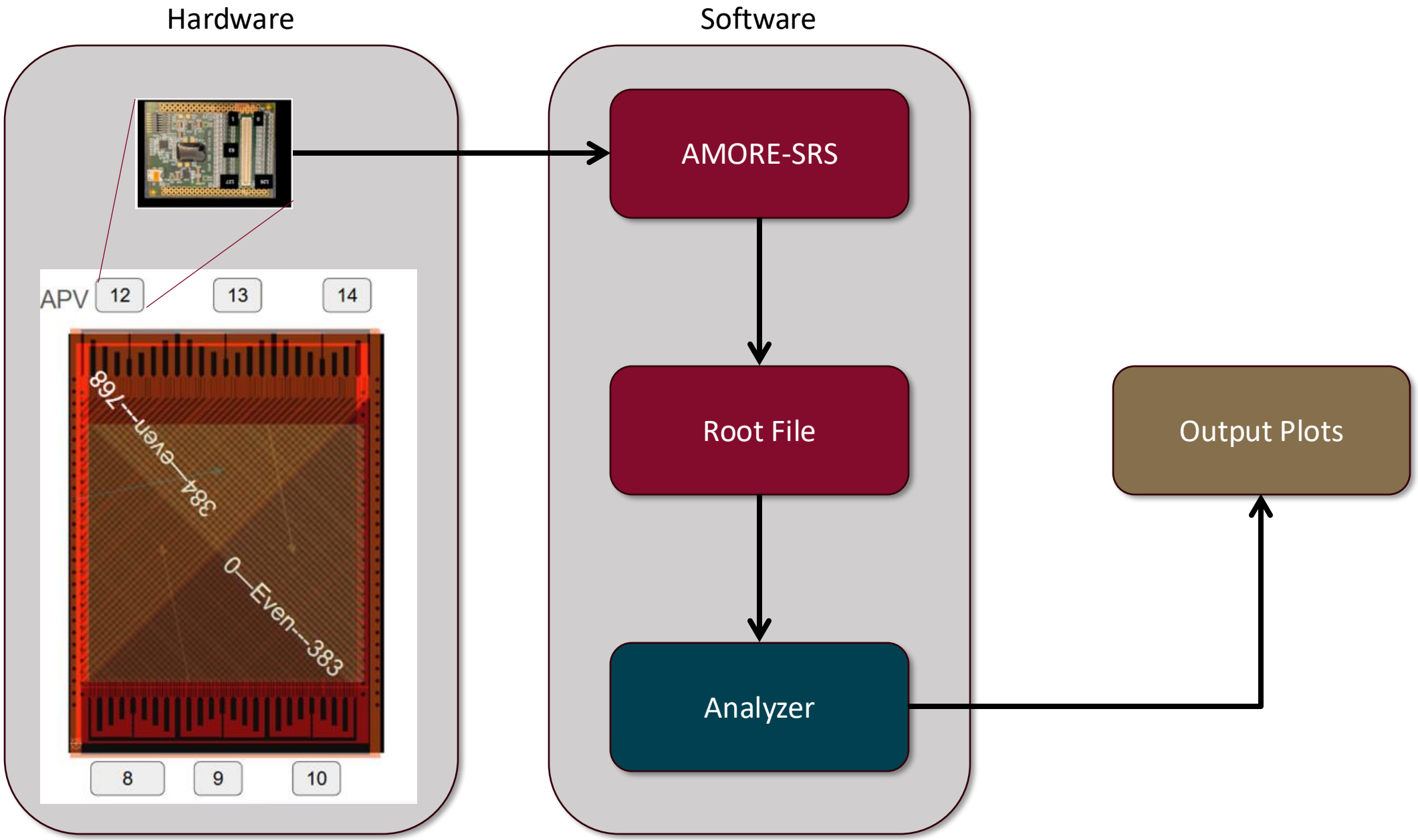
- **Detector-** A detector is a device used in particle physics to measure the position of charged particles.
- **µRWELL (Cylindrical micro-resistive well detector)-** µRWELL is a type of micro-pattern gaseous detector that combines gas electron multiplication with a resistive protection layer and a segmented readout structure to measure charged particles with high precision [1]. Designing a detector that is cylindrical is the novelty here.
- **Tracker-** In the context of this project, a tracker refers to an external planar detector. Four trackers are used to create a reference system for evaluating the efficiency of the cylindrical detector.
- **Readout board-** A readout board is an electronic circuit board that collects signals from detector channels. The readout board hosts APV chips and interfaces between the detector strips and the data acquisition system. Each readout board can only capture a single plane of the data. So in planar detectors, typically there is one for x and y positions, respectively.
- **Channels/strips-** Channels or strips are the readout elements on the detector surface that collect charge from ionization events. Each strip is connected to a specific APV channel. Strip signals are used to reconstruct particle hit positions.
- **APV chip (Analog Pipeline Voltage chip)-** An APV chip samples analog signals from each channel on the readout board in time bins.
- **Time bin-** A time bin is a discrete time sample recorded by the APV chip. Typically, 12-16 time bins are used to reconstruct the pulse shape of a beam.
- **ADC (Analog to Digital Converter value)-** ADC represents a numerical value for the current recorded on a particular strip.

Definitions cont.

- **AMORE-SRS (Analysis and Monitoring Of Readout Electronics and Scalable Readout System)-** AMORE-SRS is a live data acquisition framework for monitoring and organizing the data from the APV chips into a root file for analysis.
- **Root file-** The root file stores event data such as strip information, ADC values, and metadata. The data is organized into branches that correspond to specific variables.
- **3D Pulse plot-** A 3D pulse plot is a visualization of a single event where the x axis is time bin, the y axis is the strip number, and the z axis is the ADC value.
- **Resolution plot-** A resolution plot is a reconstruction of the 2d hit positions represented as a heat map.
- **Efficiency plot-** Efficiency plots compare the detectors efficiency against other performance characteristics such as amplification or drift field. Efficiency itself is defined as the ratio of the number of events with strip clusters seen in all five detectors divided by the number of events with strip clusters seen in just the four trackers [2].

Channel Orientations




Hardware
Software
APV
12
13
14
384—Even—768
0—Even—383
8
9
10
AMORE-SRS
Root File
Analyzer
Output Plots